Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/20/2006
PI Name:	Chelliah, H. K. Ph.D.		
Project Title:	Heterogeneous Combustion of Solid Fuel Particles Under Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	22903-3320	**Congressional District:**	5
Comments:	changed email address 10/13/06		
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	10/01/2003	**End Date:**	09/30/2006
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04AA30A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Heterogeneous combustion of solid fuels provides a substantial fraction of our energy needs here on earth, as well as in space propulsion applications. An effective and common method of solid fuel combustion is in its particulate form, however, the current theoretical/numerical models available to describe the heterogeneous combustion of such solid particles lack the complex coupling of physical-chemical processes involved. Furthermore, well designed and reliable experimental data are lacking to validate any model development effort. This is certainly true for (a) porous char/graphite particle combustion, which is relevant to efficient and clean combustion of char and (b) magnesium (Mg) particle combustion in CO2/CO environment, applicable for propulsion and energy utilization in Martian atmosphere. The goal of the proposed work is to establish a comprehensive theoretical and experimental approach utilizing NASA Microgravity facilities to develop and validate heterogeneous combustion submodels. The proposed effort will significantly enhance our understanding of the condensed fuel utilization processes, develop		

	accurate modelling approaches for a wide range of fuels, facilitate future exploration of the Martian atmosphere, and educate young scientist about fundamental concepts of combustion.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Enhance our understanding of the condensed fuel utilization processes, develop accurate modelling approaches for a wide range of fuels, facilitate future exploration of the Martian atmosphere, and educate young scientist about fundamental concepts of combustion.
Task Progress:	No progress report received this reporting period.
Bibliography Type:	Description: (Last Updated:)